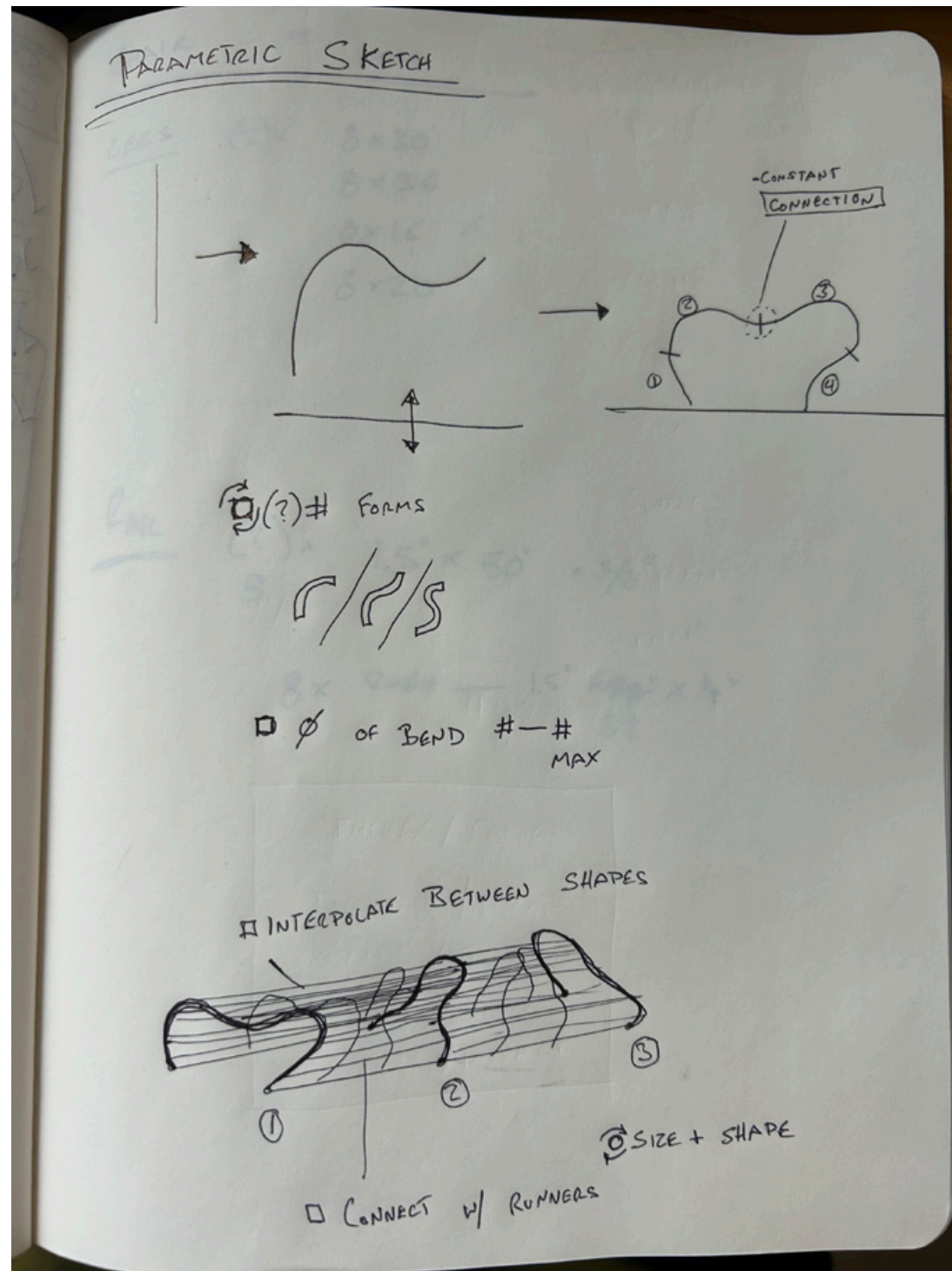


ASSIGNMENT 02
PAVILION
ARCH 565 – Advanced Computer Applications
Pierce Morley

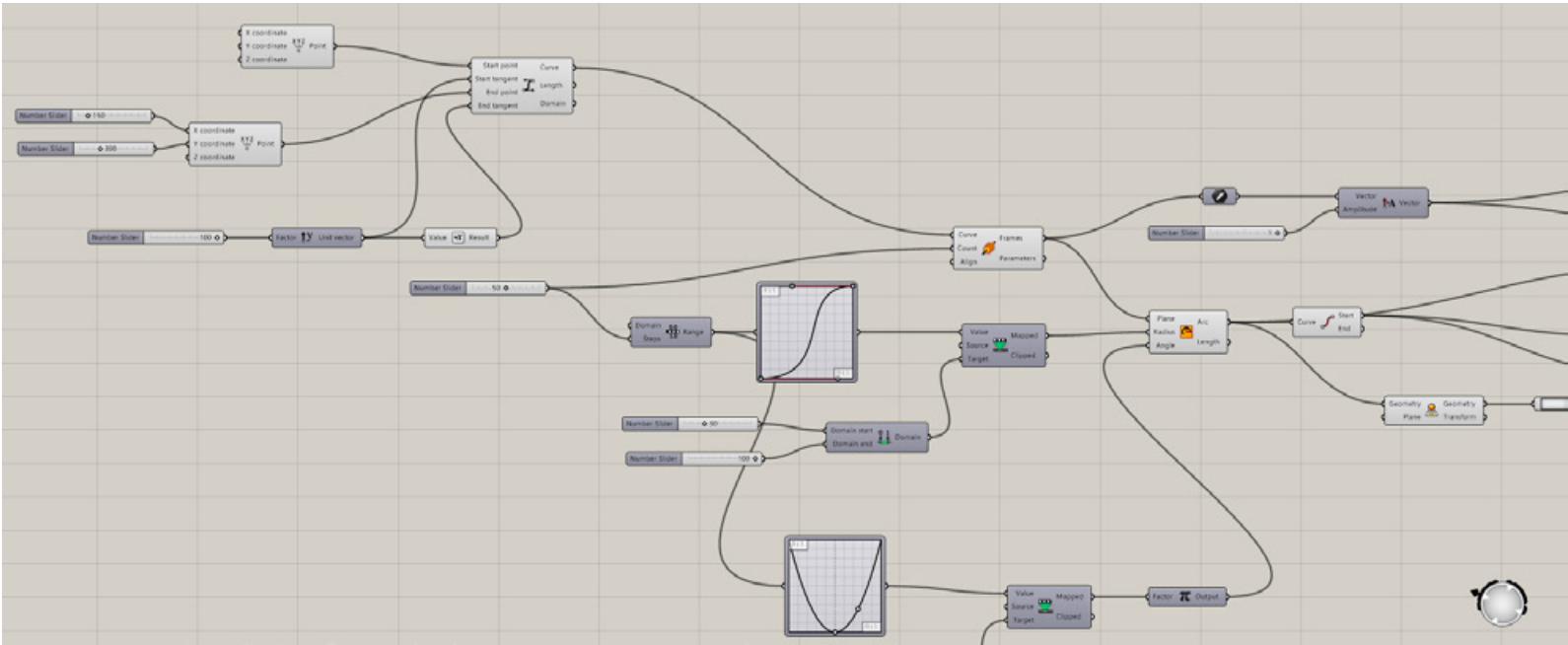
1. Parametric Sketch



This idea comes from my fourth-year studio, which focused on showcasing local boatbuilding heritage in Turkey. Drawing from those methods, the structure is constructed as largely prefabricated in the shipyard similar to a boat hull where primary beams are bent into large glulam ribs.

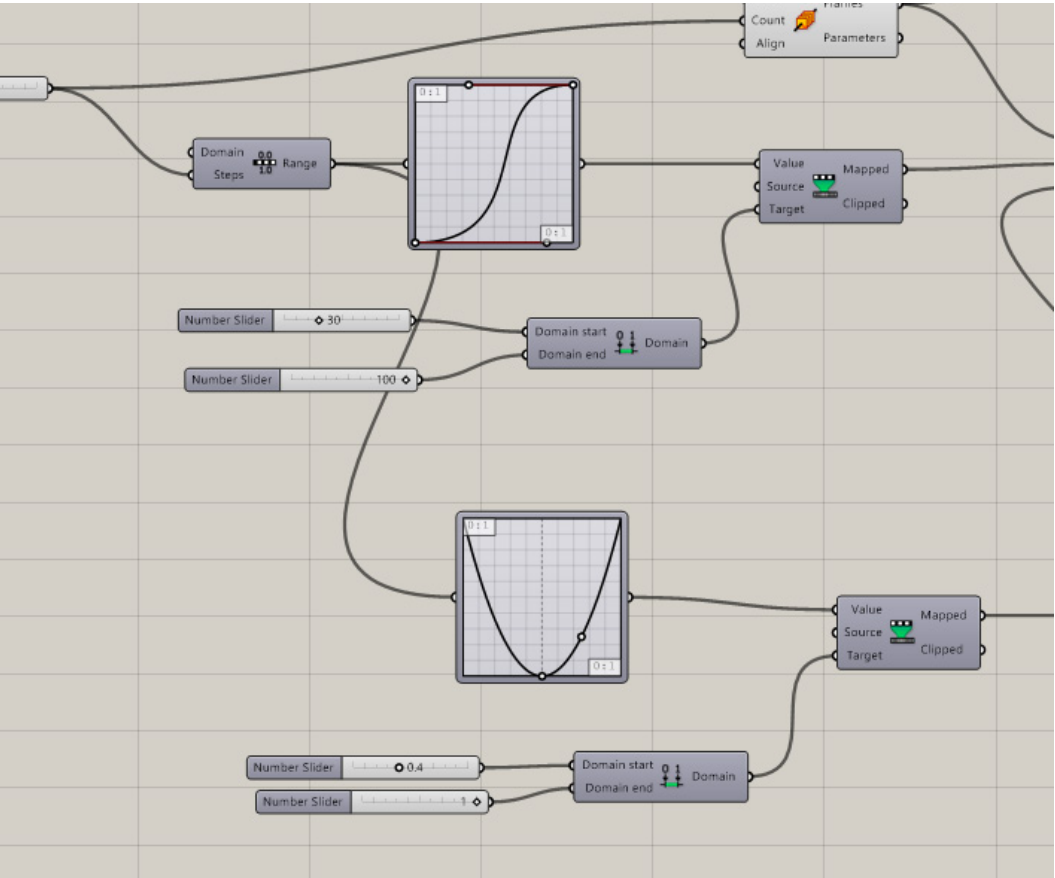
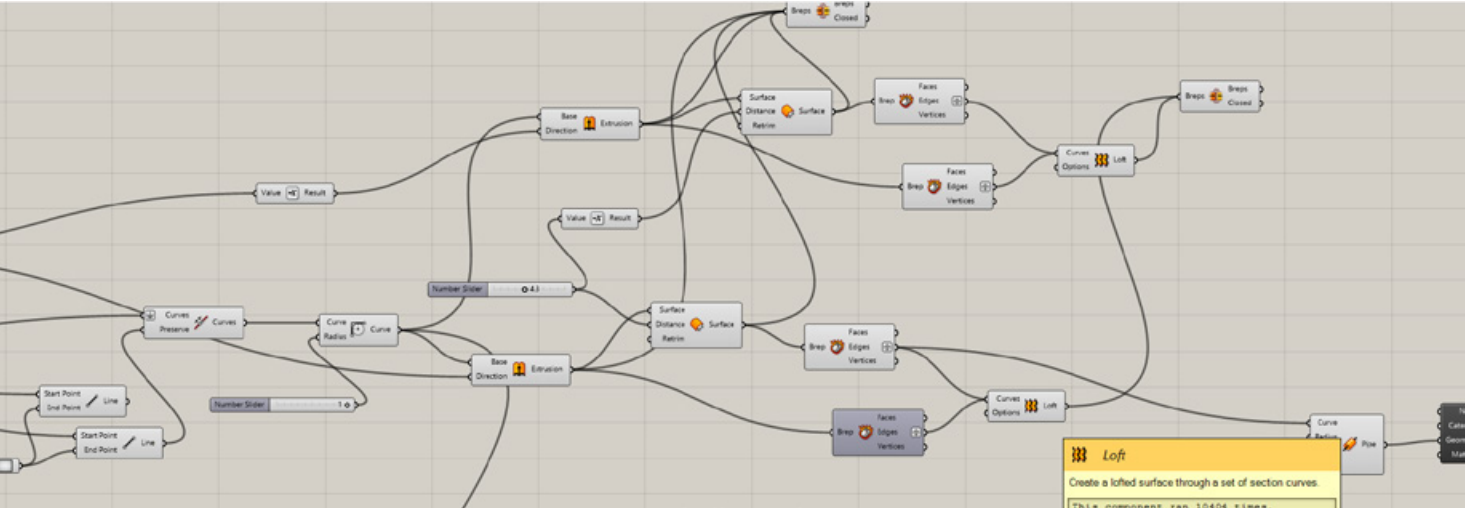
The approach aims to limit the number of formworks and reduce material waste, while still allowing for a wide range of spatial and formal variation.

2. Modeling



Grasshopper Script

Using Grasshopper in Rhino 8, I attempted to generate a structure based on my concept sketch and the parameters I set. Since I haven't used Grasshopper much before, there was a definite learning curve, and I struggled to fully achieve the form I initially imagined. That said, after some experimentation, it was rewarding to see a structure emerge from my ideas. The ability to quickly adjust and iterate through control parameters in Grasshopper rather than manually remodeling in Rhino proved especially valuable.



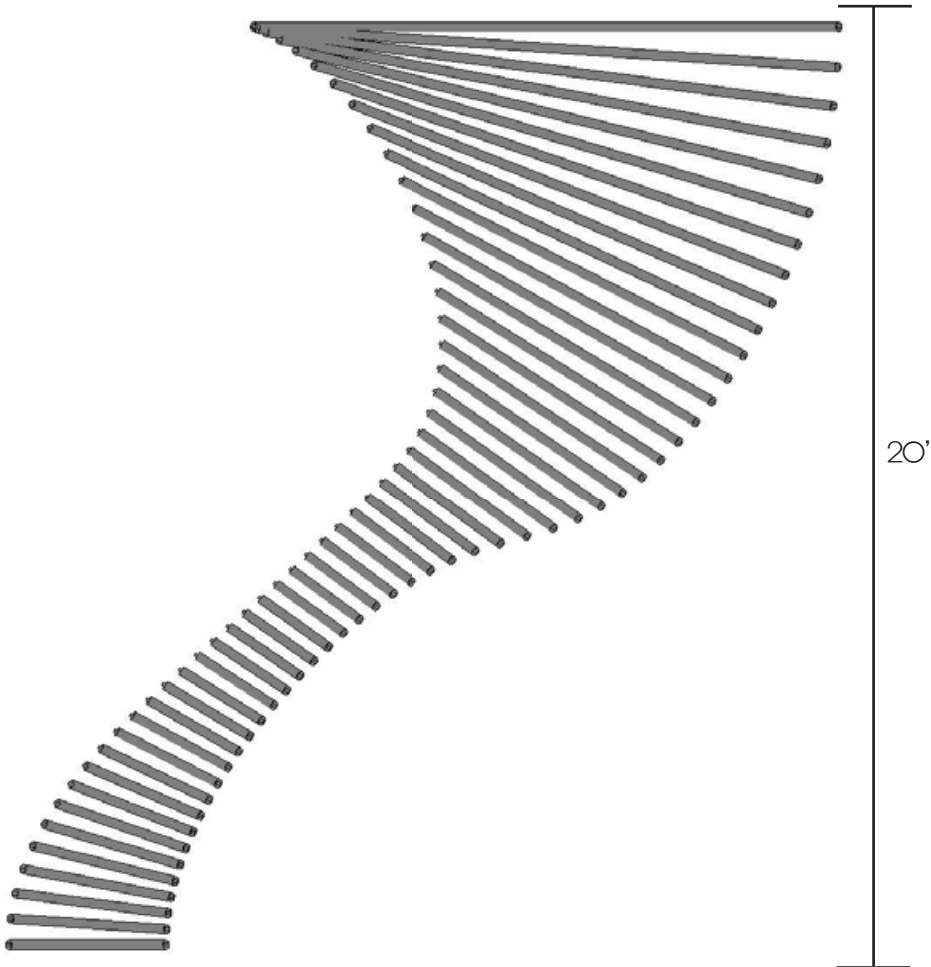
Control Parameters

3. Documentation

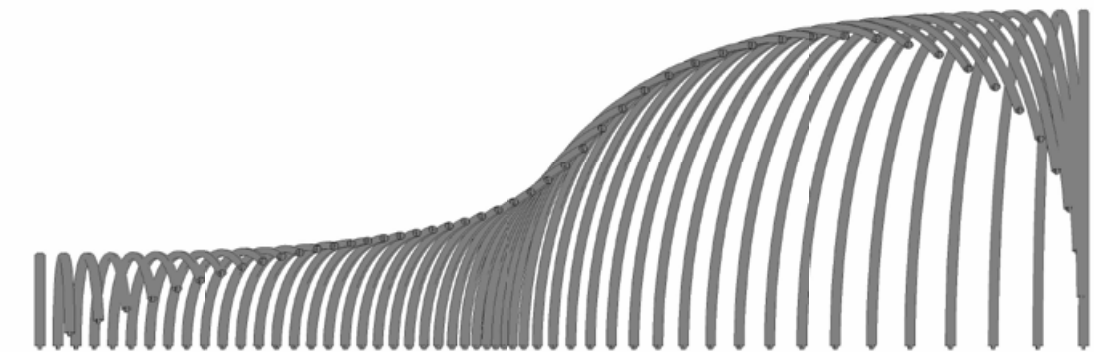
Revit with Rhino.Inside was used to document my parametric pavilion design. Because the pavilion is made from a consistent material system, I found it most effective to pipe the curves and bring them in as a DirectShape. From there, the documentation drawings could be produced in Revit pretty seamlessly, just like a typical workflow.



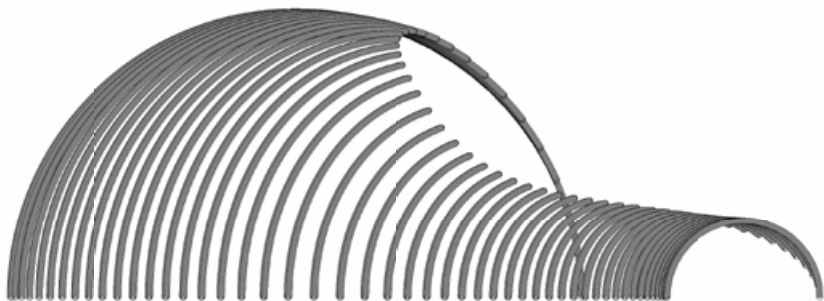
Perspective View



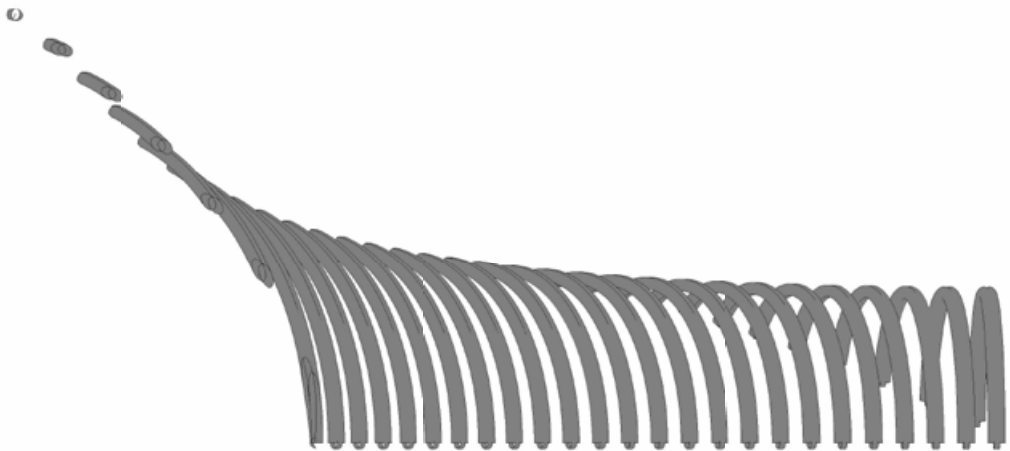
Site View



East Elevation



South Elevation



West Section



DirectShape Command

4. Rendering

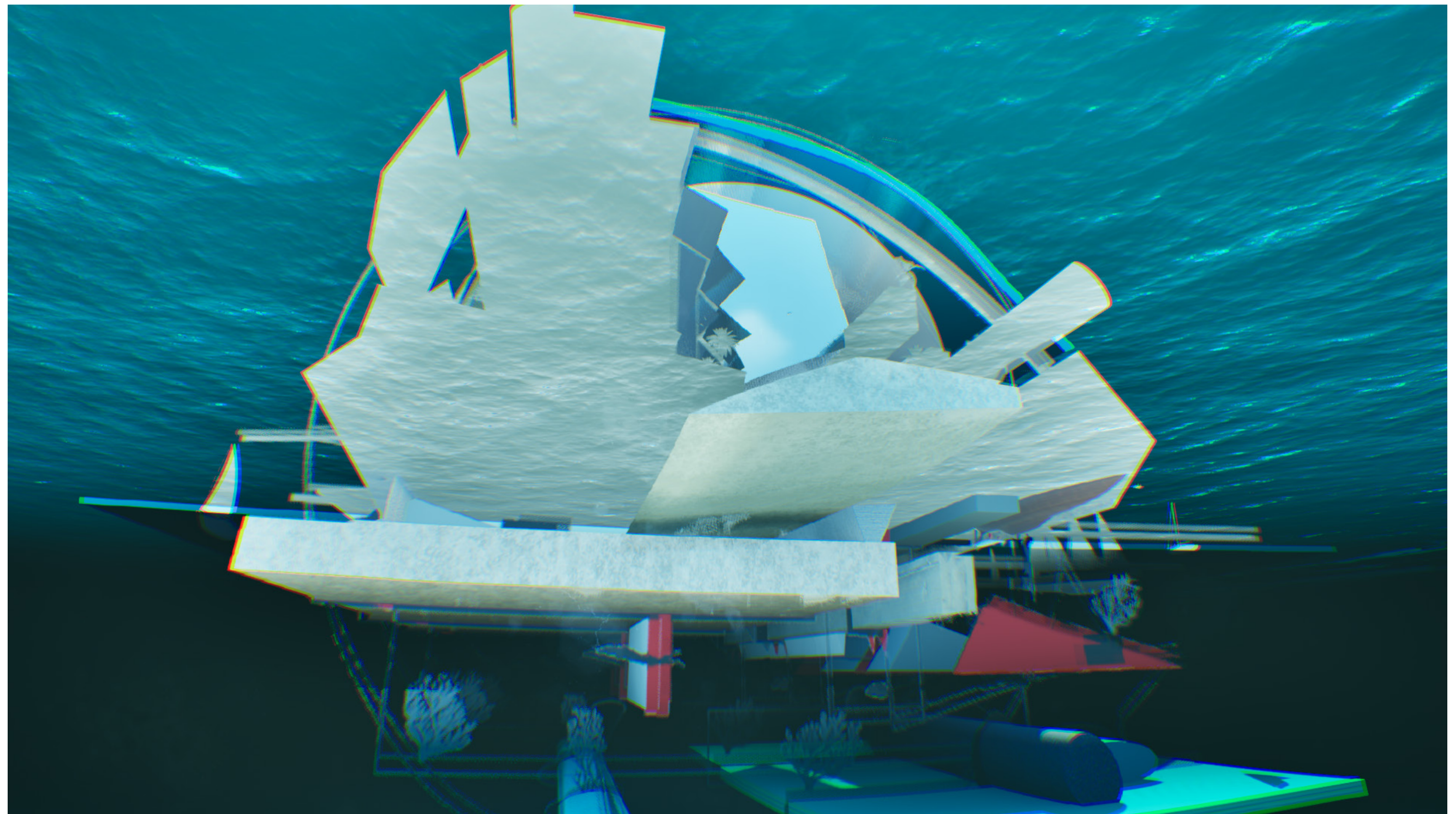
Twinmotion was used as the primary rendering software alongside my current studio project for this assignment. Though it was easy to import my Revit model, and I appreciated how seamlessly the interoperability tools synced, allowing for near-instant updates across platforms.

The project itself is a conceptual model, aimed at capturing some of the experiential qualities of coastal architecture.

I've worked with a range of rendering software, but Twinmotion stands out as one of the most intuitive and accessible. It was especially valuable to be able to adjust "camera" settings and maintain a high level of control over the surrounding environment. More than anything, it shifted how I think about rendering, less as a purely technical output, and more as the composition of an image or photograph.



Exterior Perspective



Underwater Perspective



Interior Perspective



Birds Eye View